

Procedure for soils

Preparation

1. If soils are moist, dry them up @ 40°C
2. Crush sediments if necessary (although it may break long phytoliths)
3. Sieve the sediments with 500 μ mesh
4. Weight. 10 gr., and divide the sample in 2x5gr, put sediments in 50ml centrifuge tube.
5. Add dH₂O, shake, and leave 10 minutes
6. Test pH

Carbonate digestion

8. Eliminate carbonates with HCl @ 1M in a hot bath, then with full strength HCl @ 37% if no reaction. If reaction is too strong and overflowing is possible, use ethanol to break up surface tension of bubbles/foam. Prepare also one or two beakers in case of emergency, to pour out the reacting liquid if it threatens to overflow.
9. Dilute > 50ml with dH₂O
10. Centrifuge 10min @2500rpm

Organic Digestion

11. Add with a pipette full strength HNO₃ @ 65-67%. Check for reaction.
12. Then put in hot bath and check for reaction. If too strong take it out. Prepare one or two beakers in case of emergency, to pour out the reacting liquid if it threatens to overflow. Stir from time to time. Leave until no more reaction is visible
13. Possibility to add a pinch of KClO₃ to speed up the reaction
 - a. If KClO₃ has been added, dilute > 50ml with dH₂O and rinse 2x with H₂O. Centrifuge 10min @2500rpm each time between the steps
 - b. If the breaking of metallic bonds and/or organic colloid removal and/or extract clarification steps are skipped, dilute > 50ml with dH₂O. Centrifuge 10min @2500rpm. rinse 2 times with dH₂O, centrifuge each time 10min @2500rpm.

Breaking metallic bonds (optional)

Note: this step is necessary for tropical soils containing metal complexes (degradation), but less so for other soils unless they are very clayey. Otherwise, adding NHO₃ directly after HCL can have the same effect on other sediments.

14. Add with a pipette a 1:1 mix of full strength of HNO₃-HCl (although normally Aqua Regia is ~3:1 HCL NHO₃ see annex for exact calculations, preparation and precautions)
15. Put in hot bath until the end of the reaction
16. Dilute > 50ml with dH₂O, and rinse 2x with dH₂O. Centrifuge 10min @2500rpm

Eliminating organic colloids (optional)

If soils had a lot of organic matter or are very humic

17. Add 30ml KOH @ 10% and put in hot bath for 5 min max
18. Dilute > 50ml with dH₂O, and rinse 2x with dH₂O. Centrifuge 10min @2500rpm

Clearing of extract (optional)

19. Possibility to clear up the extract if brown or deep orange with 20ml household bleach but no more than 5 minutes. Then dilute >50ml with dH₂O, centrifuge 10min @2500rpm and rinse 2x.
20. Dilute > 50ml with dH₂O, and rinse 2x with dH₂O. Centrifuge 10min @2500rpm

Deflocculation

21. Put sediments in Erlenmayer together with 2,5-5ml sodium bicarbonate (NaHCO₃) or disodium dihydrate EDTA 5%. Put in shaker for a couple of hours (overnight is best)

Elimination of silts

22. Erlenmayers are emptied in beakers of 1000ml, then 10cm of dH₂O @ 22-25°C is added. Shake well and let decant for 1h30. Siphon up to 1.5 cm. Repeat the whole operation until water is clear.
23. Concentrate in 50ml tubes used previously for 10min @2500rpm

Flotation & recovery of phytoliths

24. Add 20ml of avec sodium polytungstate @ 2.35 sg., shake well, and centrifuge 10min @ 2500rpm
25. Pipet off the surface in a 15ml centrifuge tube.
26. Repeat operation 21-22 if necessary
27. Dilute with dH₂O, centrifuge 10 min @3000rpm
28. Discard 2/3 of supernatant
29. Rinse 2x with dH₂O, centrifuge 10 min @3000rpm but discard then all the supernatant

Storage

30. Transfer in Eppendorf 1,5ml tubes. Acetone can be used to pipet off correctly.
31. Centrifuge 5min @ 3000rpm and empty it at once on a paper tissue and leave it to dry

Mounting

32. Clean the slides with ethanol and dry them up with paper
33. Take a pinch of dried extract with a glass pipette, put 1-2 drops of Canada Balsam™ or Permount™, stir slowly, spread according to the shape of the slide cover, cover and press to make the mix slightly overflow in order to avoid it to shrink and create empty bubbles on the slide when drying.

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